

Scalability & Future Plan

Date	05 October 2026	Team ID	SWTID-2026-3533
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models		
Maximum Marks	1 Mark		

Scalability & Future Plan: This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state, along with work assignments for team members.

TEAM WORK ALLOCATION

S.No	Team Member	Assigned Module / Role	Key Responsibilities
1	C. Iswarya	Team Lead & AI Integration	Core Gemini API integration, prompt engineering for comic story & script generation, model response parsing.
2	Hasini.D	Backend & Database	Backend microservices architecture, Redis caching, AWS S3 / Cloudflare asset storage, and database design.
3	Ramya.A	Frontend & UI/UX	Comic panel layouts design, canvas editor, character styling UI, and responsive frontend implementation.
4	Nandhini	Performance & Export	Celery asynchronous task workers, WebSockets setup, high-resolution PDF/image export engine.
5	Nivetha.K	Security & QA	OAuth2/JWT user authentication, KMS key management, rate limiting implementation, and system integration testing.

CURRENT SYSTEM LIMITATIONS

S.No	Limitation	Impact	Priority
1	Gemini API Rate Limits & Token Overhead	Causes latency during peak concurrent comic generation requests.	High
2	Local Storage of High-Res Comic Images	Limits scalability of user story archives on standard web server storage.	Medium
3	Character Visual Consistency Across 10+ Panels	Complex prompt drift occurs when story arcs exceed 8-10 consecutive panels.	High

SCALABILITY PLAN

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-instance server handling up to 50 concurrent users	Implement distributed microservices with Redis caching and AWS Auto Scaling groups.
2	Data Storage	Local disk storage for comic images and metadata	Migrate image asset storage to AWS S3/ Cloudflare R2 with CloudFront CDN distribution.
3	Performance	Synchronous Gemini API processing (~8s per panel generation)	Transition to asynchronous Celery background workers with WebSockets for real-time updates.
4	Security	Basic API key authentication & environment variables	Deploy OAuth2/JWT user authentication, API rate limiting, and KMS secret management.

FUTURE ROADMAP

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Core Gemini Comic Generator, Panel Layouts & PDF Export	Month 1-3	Functional MVP with end-to-end comic creation capabilities.
Phase 2	Multi-lingual Speech & Dynamic Voiceover Synthesis per Character	Month 4-6	Enhanced accessibility and global reach for comic creators.
Phase 3	Fine-tuned Custom Gemini LoRA Models for Specific Art Styles	Month 7-9	Superior visual artistic consistency and user creative control.
Phase 4	Interactive Web-based Collaborative Comic Creator & Publishing Platform	Month 10-12	Enterprise & community ecosystem with monetization options.

Overall Team Coordination Rating (1-5)	5/5 (Exceptional synchronization and seamless handoffs)
Any issues during demo	Minor initial network latency during live TTS playback test
How issues were resolved	Pre-generated backup audio cache was seamlessly utilized with zero downtime

Team Involvement in Demonstration

Date	: 15 March 2026	Maximum Marks	: 1 Mark
Team ID	: SWTID-2026-3533	Project Name	: ComicCraft - AI Comic Story Creator using Gemini Models

Team Involvement in Demonstration:

This document records the active participation and roles of each team member during the project demonstration. It ensures that every member contributes meaningfully to the presentation and that responsibilities are distributed fairly and clearly.

S.No	Team Member Name	Role in Demo	Section Presented	Contribution Summary	Participation
1	C.Iswarya	Team Leader & Presenter	Introduction, Problem Statement & Demo Conclusion	Led presentation flow, slide narration, project scope outline, and overall team coordination.	Active
2	Hasini.D	AI/ML Lead Demonstrator	Gemini API Architecture & Script Generation	Live demo of prompt engineering, Gemini model script pipeline, story arc structuring, and character consistency.	Active
3	Ramya.A	Frontend Lead Presenter	Comic Panel Layout & Visual Renderer	Demonstrated drag-and-drop comic strip builder, image rendering pipelines, and dialogue balloon UI components.	Active
4	Nandhini	Full Stack Dev Presenter	TTS Audio Playback & PDF Export Engine	Showcased narration voiceover audio generation, dynamic speech synchronization, and high-resolution document export.	Active
5	Nivetha.K	QA & Deployment Lead	User Workflow, Testing & System Deployment	Demonstrated complete user flow, cloud backend integration, edge-case validation, and performance benchmarks.	Active

Team Coordination Notes:

Team Leader / Coordinator	C.Iswarya (Team Coordinator & Lead Presenter)
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Team ID:

SWTID-2026-3533

Max Marks:

1 Mark

Project Name:

ComicCraft - AI Comic Story Creator using Gemini Models

Project Demo Planning: A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the project, including the flow of the demo, key features to highlight, and responsibilities assigned to each team member according to order.

DEMO SCHEDULE & WORK ALLOCATION

S.NO	DEMO SECTION	DESCRIPTION	DURATION	RESPONSIBLE MEMBER
1	Introduction & Problem Statement	Welcome, team introduction (SWTID-2026-3533), overview of comic creation challenges & ComicCraft mission.	2 mins	C. Iswarya
2	System Architecture & Gemini Integration	Explanation of Gemini 1.5 model integration, prompt pipeline, and UI/backend tech stack.	3 mins	Hasini. D
3	AI Script & Character Generation Demo	Live creation of a superhero comic script and character prompt engineering.	4 mins	Ramya. A
4	Comic Layout & Visual Panel Rendering	Rendering panels into grid layouts, balloon dialogue placement, and style customization.	4 mins	Nandhini
5	Audio Voiceover & Export Demonstration	Demonstrating TTS speech playback and exporting final comic to high-res PDF format.	3 mins	Nivetha. K

DEMO FLOW SUMMARY

STEP	ACTIVITY	NOTES
1	Introduction & Problem Statement	Setting project context, target audience, and key problem solved by ComicCraft.

STEP	ACTIVITY	NOTES
2	Solution Overview	Highlighting architecture, Gemini API usage, character consistency & layout engine.
3	Live Feature Demonstration	End-to-end live creation from prompt input to final PDF comic export.
4	Q&A Session	Addressing evaluator questions regarding API latency, cost optimization, and model fine-tuning.

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Communication Plan

Effective communication is essential for a successful project demonstration. This document outlines the communication strategy used within the team and with stakeholders throughout the project lifecycle, including how updates, issues, and feedback were managed.

S.N	Communication Type	Frequency	Channel/Tool	Participants	Purpose
1	Team Standup	Daily	Slack/GitHub Discussions	Full Team	Daily sync on comic generation pipeline, API integration & blockers
2	Progress Update	Weekly	Google Meet / Email	Team & Project Mentor	Review completed Gemini model features and UI panel prototypes
3	Issue / Bug Discussion	As Needed	GitHub Issues / Discord	Dev Team	Troubleshoot prompt engineering, API rate limits & export bugs
4	Stakeholder Review	BiWeekly	Google Meet	Team & Stakeholders	Milestone demonstration of AI comic creation workflow
5	Final Demo Rehearsal	Once	Google Meet	Full Team	Full run-through of end-to-end comic story creator presentation
6	Post-Demo Retrospective	Post-Demo	Slack/Google Meet	Full Team	Review project feedback and outline future scalability roadmap

Team Member Work Assignments

S.No	Member Name	Assigned Role	Key Responsibilities & Tasks
1	C.Iswarya	Team Lead & AI Integration	Overall project coordination, Gemini API integration, prompt engineering pipeline for story generation, and daily standup facilitation.
2	Hasini.D	Frontend Developer	Designing and developing UI/UX panel layouts for comic creation, visual storyline canvas, and export user flow.
3	Ramya.A	Backend & API Specialist	Setting up backend server workflows, handling Gemini API rate limits, asynchronous request queues, and image rendering services.

S.No	Member Name	Assigned Role	Key Responsibilities & Tasks
4	Nandhini	QA & Performance Engineer	Testing comic output consistency, bug tracking, rate-limiting stress testing, and managing GitHub issues and resolutions.
5	Nivetha.K	Documentation & Demo Lead	Preparing communication reports, technical documentation, presentation decks, and coordinating milestone demo rehearsals.

Communication Challenges & Resolutions

S.No	Challenge Faced	Resolution / Action Taken
1	Gemini API Rate Limits & Response Latency during development	Implemented asynchronous queueing, mock response fallback during local tests, and instant alert via Discord webhook.
2	Coordinating across different team member schedules	Established core asynchronous check-in hours on Slack and recorded Google Meet demo rehearsals for offline review.
3	Maintaining comic visual style consistency across story panels	Created a shared prompt engineering guide in GitHub Wiki and held dedicated UI alignment sessions.

Demonstration of Proposed Features

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Team Members	1. C.Iswarya 2. Hasini.D 3. Ramya.A 4. Nandhini 5. Nivetha.K		

Demonstration of Proposed Features:
This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented, demonstrated, and assigned to individual team members. It serves as evidence of the team's ability to deliver on the proposed solution.

S.N	Feature Name	Description	Assigned To	Status	Demo	Remarks
1	AI Story Script Generation	Generates multi-panel comic scripts, dialogues, and scene descriptions using Gemini models.	C.Iswarya	Implemented	Yes	Seamless script creation with tone & genre customization.
2	Character Consistency Engine	Maintains consistent character appearance and art style across panels using system prompts.	Hasini.D	Implemented	Yes	Sustains character visual fidelity across story arcs.
3	Automated Comic Panel Layout	Formats generated images and dialogue balloons into customizable comic strip grid layouts.	Ramya.A	Implemented	Yes	Flexible grid templates (2x2, 3x2, single strip).
4	Text-to-Speech Dialogue Playback	Adds voiceover audio narration for comic character dialogues using speech synthesis.	Nandhini	Implemented	Yes	Interactive audio playback for immersive reading.
5	Export & Sharing (PDF/CBZ)	Exports completed comic books into downloadable high-resolution PDF and digital formats.	Nivetha.K	Implemented	Yes	High-resolution PDF export with embedded styling.
6	Community Story Marketplace	Cloud repository for users to publish, browse, and remix community-created AI comic stories.	Joint Team Task	Partial	Yes	Basic cloud sharing functional; remixing in roadmap.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6 (5 Implemented, 1 Partial)
Total Features Demonstrated	6
Overall Implementation Rate (%)	100% Demonstrated (91.7% Fully Implemented)